

MTUNZINI, South Africa

WMO: 684970

Lat: 28.950S

Lon: 31.700E

Elev: 125

StdP: 14.63

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	46.9	49.3	40.2	36.7	63.5	43.1	41.2	59.3	20.6	60.8	18.4	67.3	2.1	250	0.328

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.2	95.0	76.1	91.7	75.1	88.9	74.1	79.5	88.0	78.1	86.2	77.0	84.7	11.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	143.3	82.8	76.0	136.6	81.5	74.9	131.4	80.5	43.1	88.3	41.7	87.0	40.5	84.9	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.1	18.2	16.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	41.6	82.0	2.9	1.6	39.5	83.1	37.8	84.0	36.2	84.9	34.1	86.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	71.4	77.4	77.5	76.4	72.6	68.4	64.3	64.4	67.4	69.2	70.7	72.7	76.0	(6)
	DBStd	6.88	5.03	4.29	4.83	5.00	4.33	4.18	4.46	5.76	5.98	5.77	6.05	5.49	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	231	0	0	1	4	15	62	65	38	24	13	7	2	(9)
	CDD50	7806	848	770	817	678	571	428	448	541	576	642	682	805	(10)
	CDD65	2563	383	350	353	232	121	40	48	113	150	191	240	342	(11)
	CDH74	16721	3021	2573	2387	1189	536	207	240	637	833	1041	1537	2520	(12)
	CDH80	5201	1004	769	715	331	112	31	40	200	298	347	526	828	(13)
(14) Wind	WSAvg	6.9	7.7	6.6	6.6	6.0	5.0	4.8	5.6	6.9	7.9	8.7	8.7	8.0	(14)
(15) Precipitation	PrecAvg	37.0	4.8	4.5	4.1	2.6	1.6	1.3	1.9	1.4	2.5	4.2	4.3	3.9	(15)
	PrecMax	55.1	9.4	10.4	8.1	8.3	4.5	6.8	16.1	4.4	9.9	9.2	12.3	7.9	(16)
	PrecMin	22.2	0.8	0.9	0.9	0.1	0.0	0.0	0.0	0.1	0.3	0.9	0.6	1.3	(17)
	PrecStd	8.3	2.3	2.2	1.9	1.8	1.2	1.5	3.1	1.1	1.9	2.0	2.4	1.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.9	96.9	95.7	92.6	87.4	83.6	84.5	90.8	93.8	94.7	96.4	98.2	(19)
		MCWB	77.4	77.6	76.8	73.9	69.3	63.7	62.6	65.9	68.9	73.1	74.7	78.5	(20)
	2%	DB	92.8	91.6	90.7	87.4	83.0	79.0	79.9	85.2	87.7	88.3	90.7	92.1	(21)
		MCWB	77.0	76.7	75.9	73.0	68.1	63.7	62.4	65.2	68.2	71.6	74.5	77.7	(22)
	5%	DB	88.6	87.6	86.9	83.7	79.4	75.7	76.1	80.7	82.8	83.5	85.9	87.6	(23)
		MCWB	76.4	75.7	75.3	72.2	67.8	63.3	62.2	64.2	67.3	70.3	73.2	75.5	(24)
	10%	DB	85.1	84.3	83.6	80.4	76.5	72.5	72.7	76.4	78.2	79.3	81.9	83.8	(25)
		MCWB	75.6	75.0	74.7	71.5	66.9	62.7	62.0	63.7	66.9	69.3	72.3	74.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.7	80.1	80.8	77.3	72.4	68.7	67.0	69.9	72.9	75.6	77.8	82.0	(27)
		MCDB	88.6	89.7	87.2	85.0	80.8	75.8	74.9	82.5	84.2	86.8	90.5	90.8	(28)
	2%	WB	78.8	78.4	78.3	75.2	70.4	66.7	65.4	68.0	70.7	73.3	75.7	79.1	(29)
		MCDB	87.1	86.9	85.0	82.5	78.6	73.7	73.7	78.0	80.7	84.0	85.8	88.2	(30)
	5%	WB	77.4	77.1	77.0	73.7	69.1	65.3	64.2	66.6	69.1	71.7	74.5	77.3	(31)
		MCDB	85.4	84.5	83.9	80.8	76.7	72.1	72.2	75.3	78.7	80.2	82.9	84.9	(32)
	10%	WB	76.3	76.0	75.6	72.4	67.9	63.9	63.0	65.2	67.9	70.3	73.1	75.7	(33)
		MCDB	83.6	82.5	81.6	78.6	74.9	70.3	70.5	73.3	75.7	77.6	80.2	82.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	14.2	14.7	15.7	18.7	20.9	20.5	18.5	16.0	14.3	14.2	14.1	(35)
		MCDBR	21.0	20.0	20.3	21.0	22.8	24.7	25.1	25.1	24.8	23.3	23.0	22.0	(36)
	5% WB	MCWBR	7.9	7.2	7.8	8.1	10.2	11.5	10.3	8.7	8.7	9.0	9.1	9.0	(37)
		MCWBR	17.4	16.5	17.6	17.7	20.0	21.0	21.1	20.2	19.8	19.2	18.8	18.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.447	0.431	0.430	0.405	0.377	0.363	0.370	0.433	0.484	0.500	0.469	0.458	(40)
		taud	2.323	2.370	2.360	2.388	2.428	2.444	2.414	2.227	2.109	2.107	2.229	2.275	(41)
	Ebn at Noon		285	284	274	264	257	253	256	251	253	261	277	283	(42)
		Edh at Noon	43	40	39	34	30	28	30	39	49	52	48	46	(43)
(44) All-Sky Solar Radiation	RadAvg		1914	1874	1634	1317	1132	1004	1070	1245	1425	1551	1743	1904	(44)
	RadStd		164	143	93	77	55	46	68	80	89	127	158	135	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page